

Exp : 5 VPS / Virtual Machine Setup and Linux Configuration

Aim

To provision a Virtual Private Server (VPS) or create a virtual machine using virtualization software such as **VirtualBox** or **VMware Workstation**, install a Linux operating system, and configure network settings including IP address, subnet mask, gateway, and DNS.

Theory

A **Virtual Private Server (VPS)** is a virtualized environment that acts like a dedicated server. It can be hosted on cloud platforms or created locally using virtualization tools. Virtualization software such as:

- VirtualBox
- VMware Workstation allows running multiple operating systems on a single physical machine. Linux distributions like:
 - Ubuntu
 - CentOS
 - Debian

are commonly used due to stability, security, and open-source nature.

Procedure

Step 1: Install Virtualization Software

Option A: Install VirtualBox

1. Download from official website
2. Install with default settings

Option B: Install VMware Workstation

1. Download installer
2. Follow installation wizard

Step 2: Create Virtual Machine In VirtualBox:

1. Open VirtualBox → Click **New**
2. Enter:
 - Name: LinuxVM
 - Type: Linux
 - Version: Ubuntu (64-bit)
3. Allocate:
 - RAM: 2 GB (minimum)
 - CPU: 1–2 cores
4. Create Virtual Hard Disk:
 - Type: VDI
 - Size: 20 GB

Step 3: Attach ISO File

1. Download ISO (e.g., Ubuntu)
2. Go to VM Settings → Storage
3. Attach ISO file

Step 4: Install Linux OS

1. Start VM
2. Select **Install Ubuntu**
3. Follow steps:
 - Language selection
 - Keyboard layout
 - Installation type (Normal)
 - Create username & password
4. Click **Install**
5. Restart VM after installation

Step 5: Configure Network (Automatic - DHCP)

Check IP: ip a

Check connectivity: ping

google.com

Step 6: Configure Static IP Address Edit

Netplan config:

```
sudo nano /etc/netplan/01-netcfg.yaml
```

Example configuration: network:

version: 2 ethernets: enp0s3:

dhcp4: no addresses:

[192.168.1.100/24] gateway4:

192.168.1.1 nameservers:

addresses: [8.8.8.8,

8.8.4.4] Apply changes: sudo

netplan apply

Step 7: Verify Network Configuration

Check IP: ip a

Check gateway:

ip route Test

DNS:

ping google.com

Step 8: (Optional) VPS on Cloud Instead

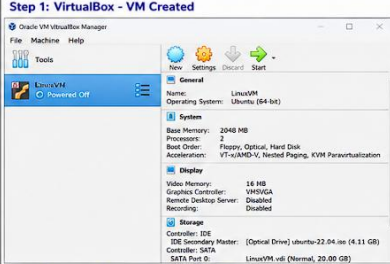
of local VM, you can use:

- Microsoft Azure
- Amazon EC2

Steps:

1. Create VM instance
2. Select Linux image
3. Configure networking
4. Connect via SSH

OUTPUT:

Step 1: VirtualBox - VM Created 	Step 2: Ubuntu Installation Completed 	Step 3: IP Address (DHCP - Automatic) <pre>student@linuxvm:~\$ ip a 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo valid_lft forever preferred_lft forever inet6 ::1/128 scope host valid_lft forever preferred_lft forever 2: enp8s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP link/ether 08:00:27:3a:1c:6b brd ff:ff:ff:ff:ff:ff inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp8s3 valid_lft 82665sec preferred_lft 82665sec inet6 fe80::a00:27ff:fe3a:1c6b/64 scope link valid_lft forever preferred_lft forever student@linuxvm:~\$</pre>
Step 4: Internet Connectivity Test (Ping Google) <pre>student@linuxvm:~\$ ping google.com PING google.com (142.250.192.78) 56(84) bytes of data: 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=117 time=23.0 ms 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=117 time=22.8 ms 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=117 time=22.8 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 2004ms rtt min/avg/max/ndev = 22.616/22.922/23.002/0.076 ms student@linuxvm:~\$</pre>	Step 5: Configure Static IP (Netplan Configuration) <pre>student@linuxvm:~\$ sudo nano /etc/netplan/01-netcfg.yaml network: version: 2 ethernet: enp8s3: dhcp4: no addresses: [192.168.1.100/24] gateway4: 192.168.1.1 nameservers: addresses: [8.8.8.8, 8.8.4.4]</pre>	Step 6: Apply Netplan Configuration <pre>student@linuxvm:~\$ sudo netplan apply student@linuxvm:~\$</pre>
Step 7: Verify Static IP <pre>student@linuxvm:~\$ ip a 2: enp8s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP link/ether 08:00:27:3a:1c:6b brd ff:ff:ff:ff:ff:ff inet 192.168.1.100/24 brd 192.168.1.255 scope global enp8s3 valid_lft forever preferred_lft forever inet6 fe80::a00:27ff:fe3a:1c6b/64 scope link valid_lft forever preferred_lft forever student@linuxvm:~\$</pre>	Step 8: Verify Gateway <pre>student@linuxvm:~\$ ip route default via 192.168.1.1 dev enp8s3 192.168.1.0/24 dev enp8s3 proto kernel scope link src 192.168.1.100 student@linuxvm:~\$</pre>	Step 9: Verify DNS (Ping Google) <pre>student@linuxvm:~\$ ping google.com PING google.com (142.250.192.78) 56(84) bytes of data: 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=117 time=22.7 ms 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=117 time=22.6 ms 64 bytes from bom12s27-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=117 time=22.5 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 2007ms rtt min/avg/max/ndev = 22.473/22.606/22.757/0.118 ms student@linuxvm:~\$</pre>

Result

Successfully created a virtual machine, installed Linux OS, and configured network settings including IP address, gateway, and DNS.